Dawei Liu

Personal Information

Name: Dawei Liu Gender: Male Date of Birth: 12th August, 1991
Place of Birth: Jinzhong, Shanxi, China
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Research Focus

Seismic signal processing, seismic inversion, tensor decomposition, time-frequency analysis and machine learning

Skills

Tensorflow, Pytorch, Python, GPU, CUDA, Matlab, and Parallel Computing

Education

- September 2018 – September 2022 **Supervisor**: Wenchao Chen
  PhD student in Information and Communication Engineering, Xi'an Jiaotong University, China.
- January 2020 – July 2022 **Supervisor**: Mauricio D. Sacchi
  Visiting PhD student in Physics, University of Alberta, Canada.
- September 2015 - July 2018 **Supervisor**: Wenchao Chen
  Master student in Electronics and Communication Engineering, Xi'an Jiaotong University, China.
- September 2009 - June 2013
  Bachelor student in Communication Engineering, Chang'an University, China.

Employment

- January 2026 - Present **Affiliation:** XJTU
  Associate Professor, Xi'an Jiaotong University, China.
- November 2023 – January 2026 **Supervisor**: Mauricio D. Sacchi
  Postdoctoral Scholar, University of Alberta, CA.
- November 2022 – October 2023 **Supervisor**: Elita Yunyue Li
  Postdoctoral Scholar, Purdue University, USA.

Peer-reviewed Papers (* stands for corresponding authors)

1. Fei Li, Ruizhe Jiao, Zhenbin Xia, **Dawei Liu***, Guanghong Du, and Yang Mu (2026). Noise suppression for CRP gathers based on Self2Self with dropout. **Journal of Applied Geophysics**, in press, 106505.

2. **Dawei Liu***, Hongzhi Yu, and Mauricio D. Sacchi (2026). SeisDPS-3D: A Diffusion Model-Based Approach for 3D Seismic Data Deconvolution. **Surveys in Geophysics**, advance online publication.

3. Qingfang Wang, **Dawei Liu***, Mauricio D. Sacchi, Xiaokai Wang, Deyu Meng, and Wenchao Chen (2026). STISR: A Stacked Tucker Implicit Seismic Reparameterization Framework for 3D Seismic Data Denoising. **Petroleum Science**, 23(9), 5384-5399.

4. Xiaokai Wang, Shengpei Xia, Chuanli Xiao, Jingrui Luo, Wenchao Chen, **Dawei Liu**, Zhaodan Liu, and Ningning Zhou (2026). Directional Characteristic Analysis of Wavefields Excited by High-Speed Train Seismic Sources. **Geophysics**, 91(4), KS73-KS87.

5. Ji Li, **Dawei Liu***, and Daniel Trad (2026). Unsupervised Seismic Data Denoising via SIREN-Guided Diffusion Model. **Geophysical Prospecting**, 74(4), e70172.

6. Zeyang Liu, **Dawei Liu***, Mauricio D. Sacchi, Jingye Li, Xiaohong Chen, and Wei Zhang (2026). Diffusion Posterior Sampling with 3D Lateral Constraints for Post-Stack Impedance Inversion. **Journal of Applied Geophysics**, 251, 106248.

7. Zeyang Liu, Jingye Li, **Dawei Liu**, Wei Zhang, Guochang Liu, and Xiaohong Chen (2026). 3D Impedance Inversion Based on a Gradient-Enhanced Diffusion Model. **Petroleum Science Bulletin**, 11(2), 415-428.

8. Ji Li, **Dawei Liu***, and Daniel Trad (2026). Unsupervised Attenuation of Aliased Ground Roll in Undersampled 3-D Land Seismic Data. **IEEE Transactions on Geoscience and Remote Sensing**, 64, 5913813.

9. Yang Mu, Ruotong Liu, Mengmeng Wu, **Dawei Liu***, and Jiaojiao Li (2026). Dual-Direction Implicit Neural Representation for Seismic Scattered Noise Attenuation. **IEEE Geoscience and Remote Sensing Letters**, 23, 7504705.

10. Yinghe Wu, Shulin Pan, **Dawei Liu**, Kai Chen, Yaojie Chen, Ziyu Qin, Shengbo Yi, and Zeyang Liu (2026). Rapid Retrieval and Classification of Passive-Source Body Wave Events Using a Convolutional Self-Attention Encoder: Application to Gas Storage Monitoring. **Geophysics**, 91(1), WA123-WA135.

11. **Dawei Liu**, Zhenyu Wang, Xiaokai Wang, and Wenchao Chen (2025). Zero-Shot Denoising for DAS-VSP Data Based on Conditional Diffusion Probabilistic Models. IEEE Transactions on Geoscience and Remote Sensing, 63, 1–11.

12. Hongzhi Yu, Xiaokai Wang, Wenchao Chen, and **Dawei Liu*** (2025). Unsupervised Diffusion Model for Seismic Deconvolution. IEEE Geoscience and Remote Sensing Letters (GRSL), 22, 1-5.

13. **Dawei Liu**, Yijie He, Xiaokai Wang, Mauricio Sacchi, Wenchao Chen, Fei Li, Juan Chen and Yang Mu (2025). CycleGAN Integration of High-Resolution Crooked Lines into 3D Seismic Volumes: Enhancing Dataset Resolution on the Loess Plateau, China. Geophysics, 90, V339-V356.

14. Alejandro Quiaro, **Dawei Liu***, and Mauricio Sacchi (2025). Non-intrusive reduced basis approximation to the solution of Helmholtz equation: The magnetotellurics case. **Geophysics**, 90, WA323-WA337.

15. **Dawei Liu**, Yijie He, Xiaokai Wang, Mauricio Sacchi, Wenchao Chen, Guanghong Du, and Mengbo Zhang (2025). From Shallow to Deep: Enhancing Seismic Resolution with Weak Supervision. **Geophysics**, 90, V223-V239.

16. **Dawei Liu**, Qingfang Wang, Nan You, Mauricio Sacchi, and Wenchao Chen (2025). Filling the Gap: Enhancing Borehole Imaging with Tensor Neural Network. **Geophysics**, 90, D71-D83.

17. Yinghe Wu, Shulin Pan, **Dawei Liu**, Kai Chen, Yaojie Chen, Ziyu Qin, Shengbo Yi, and Zeyang Liu (2025). Rapid Retrieval and Classification of Passive-Source Body Wave Events Using a Convolutional Self-Attention Encoder: Application to Gas Storage Monitoring. **Geophysics**, 90, 1-86.

18. Yanglijiang Hu, Weiwei Xu, Xiaokai Wang, **Dawei Liu**, and Wenchao Chen (2025). Adaptive dictionary identification framework and its application to sparsity-optimized harmonic noise separation. **Geophysics**, 90, V161-V177.

19. Ji Li, **Dawei Liu***, and Mauricio Sacchi (2025). Unsupervised ground roll attenuation via implicit neural representations. **Geophysics**, 90, V111-V121.

20. Wenbin Gao, **Dawei Liu***, Wenchao Chen, Mauricio D. Sacchi, and Xiaokai Wang (2025). NeRSI: Neural implicit representations for 5D seismic data interpolation. **Geophysics**, 90, V29–V42.

21. Ji Li, **Dawei Liu***, Daniel Trad, and Mauricio Sacchi (2024). Robust unsupervised 5D seismic data reconstruction on both regular and irregular grids. **Geophysics**, 89, V537-V549.

22. **Dawei Liu**, Wenbin Gao, Weiwei Xu, Ji Li, Xiaokai Wang, and Wenchao Chen (2024). 5-D seismic data interpolation by continuous representation. IEEE Transactions on Geoscience and Remote Sensing, 62, 1–11.

23. Haibo Mao, Xin Zhou, Xiaofeng Li, Long Pan, Juan Lin, **Dawei Liu**, and Xiaokai Wang (2024). Intelligent noise suppression for 3D post-stack seismic data of the Junggar Basin. Coal Geology and Exploration, 52(11), 141–150.

24. Xiaokai Wang, Shengpei Xia, Xinyue Pan, Baoli Wang, **Dawei Liu**, and Wenchao Chen (2024). The Broadband Virtual Shot Gathers Construction Based on High-Speed Train-Induced Seismic Wave. IEEE Transactions on Geoscience and Remote Sensing, 62, 1–11.

25. Yanglijiang Hu, Xiaokai Wang, Qinlong Hou, **Dawei Liu**, Xinmin Shang, Meng Zhang, and Wenchao Chen (2024). Modeling and sparsity-promoting separation of wind turbine noise in common-shot gathers. **Geophysics**, 89, V87–V101.

26. **Dawei Liu**, Mei Zhou, Xiaokai Wang, Zhensheng Shi, Mauricio D. Sacchi, Wenchao Chen, Zhaodan Liu, and Xian Wang (2024). Enhancing ground penetrating radar (GPR) data resolution through weakly supervised learning. IEEE Transactions on Geoscience and Remote Sensing, 62, 1–13.

27. Xiaokai Wang, **Dawei Liu**, Wenchao Chen, and Chun Li (2024). A cascaded synchrosqueezing transform for precise analysis of seismic signals. IEEE Transactions on Geoscience and Remote Sensing, 62, 1–12.

28. Xiaokai Wang, Chunmeng Cui, **Dawei Liu**, Pu Liu, Zhensheng Shi, and Wenchao Chen

(2024). Seismic data separation based on the equidistant-spectral constrained morphological component analysis. IEEE Transactions on Geoscience and Remote Sensing, 62, 1–11.

29. Ji Li, Daniel Trad, and **Dawei Liu*** (2024). Robust seismic data denoising via self-supervised deep learning. **Geophysics**, 89, V437–V451.

30. Ji Li and **Dawei Liu*** (2024). Robust multi-dimensional reconstruction via group sparsity with Radon operators. **Geophysics**, 89, V219–V230.

31. **Dawei Liu**, Mauricio D. Sacchi, Xiaokai Wang, and Wenchao Chen (2023). Unsupervised deep learning for ground roll and scattered noise attenuation. IEEE Transactions on Geoscience and Remote Sensing, 61, 1–17.

32. **Dawei Liu**, Wenli Niu, Xiaokai Wang, Mauricio D. Sacchi, Wenchao Chen, and Cheng Wang (2023). Improving vertical resolution of vintage seismic data by a weakly supervised method based on CycleGAN. **Geophysics**, 88, 1–103.

33. **Dawei Liu**, Wei Wang, Xiaokai Wang, Zhensheng Shi, Mauricio D. Sacchi, and Wenchao Chen (2023). Improving sparse representation with deep learning: A workflow for separating strong background interference. **Geophysics**, 88, WA253–WA266.

34. Xiaokai Wang, Siyuan Fan, Chen Zhao, **Dawei Liu**, and Wenchao Chen (2023). A self-supervised method using Noise2Noise strategy for denoising CRP gathers. IEEE Geoscience and Remote Sensing Letters, 20, 1–5.

35. **Dawei Liu**, Xiaokai Wang, Xiaohai Yang, Haibo Mao, Mauricio D. Sacchi, and Wenchao Chen (2022). Accelerating seismic scattered noise attenuation in OVT domain: Application of deep learning. **Geophysics**, 87, V505–V519.

36. **Dawei Liu**, Mauricio D. Sacchi, and Wenchao Chen (2022). Efficient tensor completion methods for 5-D seismic data reconstruction: Low-rank tensor train and tensor ring. IEEE Transactions on Geoscience and Remote Sensing, 60, 1–17.

37. **Dawei Liu**, Haoqi Zhang, Xiaokai Wang, Wenchao Chen, Zhensheng Shi, and Zhonghua Zhao (2022). Separation of seismic multiple reflection-refraction based on morphological component analysis with high-resolution linear Radon transform. **Geophysics**, 87, V367–V379.

38. **Dawei Liu**, Xiangfang Li, Wei Wang, Xiaokai Wang, Zhensheng Shi, and Wenchao Chen (2022). Eliminating harmonic noise in vibroseis data through sparsity-promoted waveform modeling. **Geophysics**, 87, V183–V191.

39. Weiwei Xu, Yanhui Zhou, **Dawei Liu**, Xiaokai Wang, and Wenchao Chen (2022). Seismic intelligent deblending via plug-and-play method with blended CSGs trained deep CNN Gaussian denoiser. IEEE Transactions on Geoscience and Remote Sensing, 60, 1–13.

40. Xiaokai Wang, Zhizhou Huo, **Dawei Liu**, Weiwei Xu, and Wenchao Chen (2022). A common-reflection-point gather random noise attenuation method based on the synchrosqueezing wavelet transform. Interpretation, 10, SA59–SA67.

41. Xiaokai Wang, **Dawei Liu***, and Wenchao Chen (2022). Accelerating seismic dip estimation with deep learning. IEEE Geoscience and Remote Sensing Letters, 19, 1–5.

42. **Dawei Liu**, Lei Gao, Xiaokai Wang, and Wenchao Chen (2021). A dictionary learning

method with atom splitting for seismic footprint suppression. **Geophysics**, 86, V509–V523.

43. Yanglijiang Hu, **Dawei Liu**, Xiaokai Wang, Zhonghua Zhao, and Wenchao Chen (2022). Attenuation of the multiple reflection-refraction in 2D common-shot gathers via random-derangement-based f-x Cadzow filter. IEEE Geoscience and Remote Sensing Letters, 19, 1–5.

44. **Dawei Liu**, Zheyuan Deng, Cheng Wang, Xiaokai Wang, and Wenchao Chen (2022). An unsupervised deep learning method for denoising prestack random noise. IEEE Geoscience and Remote Sensing Letters, 19, 1–5.

45. Wenchao Chen, **Dawei Liu**, Xinjian Wei, Xiaokai Wang, Dewu Chen, Shuping Li, and Dong Li (2021). Unsupervised noise suppression method for depth network seismic data based on prior information constraint. Coal Geoplogy & Exploaration, 49(1), 249-256.

46. **Dawei Liu**, Wei Wang, Xiaokai Wang, Cheng Wang, Jiangyun Pei, and Wenchao Chen (2020). Poststack seismic data denoising based on 3D convolutional neural network. IEEE Transactions on Geoscience and Remote Sensing, 58(3), 1598–1629.

Meeting Abstracts

1. **Dawei Liu**, Mauricio D. Sacchi, Yijie He, Xiaokai Wang, Wenchao Chen, Fei Li, Juan Chen, and Yang Mu, (2025), "Improving 3D Seismic Resolution in the Loess Plateau: Leveraging 2D Crooked-Line Gully Survey through Weak Supervision," Abstract, AAPG/SEG Annual Meeting (Paper No. 4312537).

2. **Dawei Liu**, Xiaohai Yang, Xiaokai Wang, Haibo Mao, Mauricio D. Sacchi, and Wenchao Chen, (2021), "Deep learning for prestack strong scattered noise suppression," SEG Technical Program Expanded Abstracts : 1601-1605.

3. Haoqi Zhang, **Dawei Liu**, Xiaokai Wang, and Wenchao Chen, (2021), "Attenuation of multiple reflection-refraction in tau-p domain via morphological component analysis," SEG Technical Program Expanded Abstracts : 2974-2978.

4. Qinlong Hou, **Dawei Liu**, Xiaokai Wang, and Wenchao Chen, (2021), "Adaptive DAS coupling noise suppression based on local MCA," SEG Technical Program Expanded Abstracts : 2979-2983.

5. Chen Zhao, Li Jiang, Xiaokai Wang, **Dawei Liu**, Zhensheng Shi, and Wenchao Chen, (2021), "Prestack seismic noise attenuation based on 3D CWT," SEG Technical Program Expanded Abstracts : 2834-2838.

6. **Dawei Liu**, Wenchao Chen, Mauricio D. Sacchi, and Hongxu Wang, (2020), "Should we have labels for deep learning ground roll attenuation?," SEG Technical Program Expanded Abstracts : 3239-3243.

7. **Dawei Liu**, Zheyuan Deng, Xiaokai Wang, Wei Wang, Zhensheng Shi, Cheng Wang, and Wenchao Chen, (2020), "Must we have labels for denoising seismic data based on deep learning?," SEG Global Meeting Abstracts : 31-35.

8. **Dawei Liu**, Xiaokai Wang, Zhensheng Shi, Yanhui Zhou, and Wenchao Chen, (2019), "A convolutional neural network for seismic dip estimation," SEG Technical Program Expanded Abstracts : 2634-2638.

9. **Dawei Liu**, Xiaokai Wang, Wenchao Chen, Yanhui Zhou, Wei Wang, Zhensheng Shi, Cheng Wang, and Chunlin Xie, (2019), "3D seismic waveform of channels extraction by artificial intelligence," SEG Technical Program Expanded Abstracts : 2518-2522.

10. **Dawei Liu**, Wei Wang, Wenchao Chen, Xiaokai Wang, Yanhui Zhou, and Zhensheng Shi. Random noise suppression in seismic data: what can deep learning do? [C].2018 SEG Annual Meeting, 2016-2020. [*Cited by Professor Öz Yilmaz in his new book: Land seismic case studies for near-surface modeling and subsurface imaging, 2021.]*

11. Fen Zhang, **Dawei Liu**, Xiaokai Wang, and Wenchao Chen. Random noise attenuation method for seismic data based on deep residual network[C]. 2018 CPS/SEG Annual Meeting, 2018.

12. Siqi Chi, Wenchao Chen, Lu Zhang, and **Dawei Liu**, Jianyou Chen. Three-dimensional seismic texture attributes analysis based on removed strong background noise[C]. 2018 CPS/SEG Annual Meeting, 2018.

13. Jianyou Chen, Wenchao Chen, Xiaokai Wang, and **Dawei Liu**. The DAS coupling noise removal using alternating projection iteration with united sparse transforms. 2018 CPS/SEG Annual Meeting, 2018.

14. Jianyou Chen, Yuefeng Pang, Wenchao Chen, Lei Gao, and **Dawei Liu**. The analysis of space dimensionality reduction error in SVD filtering algorithm with application to VSP wavefield separation [C].2018 CPS/SEG Annual Meeting, 2018.

15. **Dawei Liu** et al., (2026), "CTR-Net: Continuous Tensor Ring Network for Off-the-Grid 5D Data Reconstruction," CIML Workshop (OpenReview).

Service to the profession

1. Associate editor for *Goephysics*
2. Session Chair for IMAGE INT session (2024)
3. Session Chair for IMAGE MLDA session (2023)
4. Guest editor for *Frontiers in Earth Science*
5. Peer Reviewer for:
   Geophysics, IEEE Transactions on Geoscience and Remote Sensing, Geoscience and Remote Sensing Letters, Acta Geophysics, Petroleum Science, International Conference on Physics, Mathematics and Statistics, Frontiers, Journal of Geophysics and Engineering, IMAGE, Digital Signal Processing

Awards and Honors

1. **Plenary and Keynote speech,** 2nd Intelligent Geophysics Annual Conference & 5th International Conference on Mathematical Geophysics, Harbin, China, 2024. Title: “High-Dimensional Seismic Data Denoising and Interpolation: From Supervised to Unsupervised.”
2. **Outstanding Presentation Award**, 4th National Geoscience Graduate Forum, 2024. Awarded for the paper “LTFRSI: Low-Rank Tensor Function Representation for Seismic Interpolation.”
3. **Best Paper Award**: Second Intelligent Geophysics Annual Conference & Fifth International Conference on Mathematical Geophysics, Harbin, China, 2023. Recognized

for contributions to high-dimensional seismic data denoising and interpolation using weakly and unsupervised learning frameworks.

4. **Outstanding Oral Presentation Award**, 2021 SEG 4th International Workshop on Mathematical Geophysics (Traditional & Learning), Beijing (virtual). Awarded for the paper "A Weakly Supervised Method for Improving the Resolution of Seismic Data Based on Cycle Generative Adversarial Network."
5. National Award scholarship for PhD Student at Xi'an Jiaotong University, October 2020
6. School outstanding postgraduate Cadre at Xi'an Jiaotong University, September 2016
7. School outstanding graduate students at Chang'an University, September 2013

Funds

1. Xi'an Jiaotong University Young Talent Start-up Fund, Principal Investigator, ¥1,000,000 RMB, 2026-2032.
2. National Science and Technology Major Project of China (Subtopic) – Seismic Prediction of Dual Sweet Spots in Unconventional Reservoirs Integrating Geological and Engineering Constraints, Principal Investigator, ¥2,800,000 RMB, 2026-2031.